

A First Course in Causal Inference Ch.4

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Ch.4 Neymanian Repeated Sampling Inference in CRE

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- ▶ Neyman (1923) has proposed an unbiased point estimator and a conservative confidence interval based on the sampling distribution of the point estimator
- ▶ This chapter covers Neyman (1923)'s fundamental parts.

4.1 Finite population quantities

- ▶ Setup: CRE with n units (treatment group: n_1 units, control group: n_0 units); For $i = 1, \dots, n$, potential outcomes $Y_i(1)$ and $Y_i(0)$ and individual effect $\tau_i = Y_i(1) - Y_i(0)$.
- ▶ We then obtain finite population means:

$$\bar{Y}(1) = \frac{1}{n} \sum_{i=1}^n Y_i(1), \quad \bar{Y}(0) = \frac{1}{n} \sum_{i=1}^n Y_i(0),$$

- ▶ variances:

$$S^2(1) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\}^2, \quad S^2(0) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(0) - \bar{Y}(0)\}^2,$$

- ▶ and covariance:

$$S(1, 0) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\} \{Y_i(0) - \bar{Y}(0)\}.$$

4.1 Finite population quantities

- The individual effects have mean

$$\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \bar{Y}(1) - \bar{Y}(0),$$

- and variance

$$S^2(\tau) = \frac{1}{n-1} \sum_{i=1}^n (\tau_i - \tau)^2$$

We then have

Lemma 4.1

$$2S(1, 0) = S^2(1) + S^2(0) - S^2(\tau).$$

4.2 Neyman (1923)'s theorem

- ▶ We are interested in estimating the average causal effect τ based on the (observed) data $(Z_i, Y_i)_{i=1}^n$ from a CRE.
- ▶ Based on the observed data, the sample means:

$$\hat{Y}(1) = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_i, \quad \hat{Y}(0) = \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_i,$$

- ▶ and the sample variances:

$$\hat{S}^2(1) = \frac{1}{n_1 - 1} \sum_{i=1}^n Z_i \{Y_i - \hat{Y}(1)\}^2, \quad \hat{S}^2(0) = \frac{1}{n_0 - 1} \sum_{i=1}^n (1 - Z_i) \{Y_i - \hat{Y}(0)\}^2.$$

- ▶ Obviously, there are no sample covariance and $S^2(\tau)$ because **we cannot observe both potential outcomes for each unit**.

4.2 Neyman (1923)'s theorem

Theorem 4.1

1. the difference-in-means estimator $\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)$ is unbiased for τ :

$$\mathbb{E}[\hat{\tau}] = \tau.$$

2. $\hat{\tau}$ has variance

$$\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} = \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1, 0).$$

3. the variance estimator $\hat{V} = \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0}$ is conservative for estimating $\text{var}(\hat{\tau})$ in the sense that $\mathbb{E}[\hat{V}] - \text{var}(\hat{\tau}) = \frac{S^2(\tau)}{n} \geq 0$ with equality holding if and only if $\tau_i = \tau$ for all units.

4.2 Neyman (1923)'s theorem

- ▶ The FRT works for any test statistic but Neyman (1923)'s theorem is only about the difference in means.
- ▶ If $S(1, 0) > 0$, then those treated units also have relatively large control potential outcomes¹. $\rightsquigarrow$ the observed outcomes under control are relatively small $\rightsquigarrow$ large $\hat{\tau}$.
- ▶ If $S(1, 0) < 0$, the those treated units have relatively small control potential outcomes². $\rightsquigarrow$ the observed outcomes under control are relatively large $\rightsquigarrow$ small $\hat{\tau}$.

¹In other words, large control potential outcomes are not “observed.”

²In other words, small control potential outcomes are not “observed.”

4.2 Neyman (1923)'s theorem

Thorem 4.2

Let $n \rightarrow \infty$ and $n_i \rightarrow \infty$. If $\frac{n_1}{n}$ has a limiting value in $(0, 1)$, $\{S^2(1), S^2(0), S(1, 0)\}$ have limiting values, and

$$\max_{1 \leq i \leq n} \frac{\{Y_i(1) - \bar{Y}(1)\}^2}{n} \rightarrow 0, \quad \max_{1 \leq i \leq n} \frac{\{Y_i(1) - \bar{Y}(0)\}^2}{n} \rightarrow 0, \quad (1)$$

then

$$\frac{\hat{\tau} - \tau}{\sqrt{\text{var}(\hat{\tau})}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (2)$$

and

$$\hat{S}^2(1) \xrightarrow{p} S^2(1), \quad \hat{S}^2(1) \xrightarrow{p} S^2(0) \quad (3)$$

4.2 Neyman (1923)'s theorem

The intuitive understanding for Thm. 4.2 is:

- ▶ Eq. (1): potential outcomes for each unit are not extremely large to dominate the variability³.
- ▶ Suppose a case one unit has an extremely large outcome, then $\hat{\tau}$ depends on which group the unit belongs to: Eq. (2) does not hold.
- ▶ Eq. (3) ensures Neyman (1923)'s conservative CI for τ , because $\hat{V} \geq \text{var}(\hat{\tau})$.

³Formally, $\max_{1 \leq i \leq n} \|Y_i(1) - \bar{Y}(1)\| = o(\sqrt{n})$.

4.3 Proofs

Unbiasedness of $\hat{\tau}$:

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_i = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_{i(0)},$$

then

$$\begin{aligned} \mathbb{E}[\hat{\tau}] &= \mathbb{E} \left[\frac{1}{n_1} \sum_{i=1}^n Z_i Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_{i(0)} \right] \\ &= \frac{1}{n_1} \sum_{i=1}^n \mathbb{E}[Z_i] Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n \mathbb{E}[1 - Z_i] Y_{i(0)} \\ &= \frac{1}{n_1} \sum_{i=1}^n \frac{n_1}{n} Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n \frac{n_0}{n} Y_{i(0)} = \frac{1}{n} \sum_{i=1}^n Y_{i(1)} - \frac{1}{n} \sum_{i=1}^n Y_{i(0)} = \tau. \quad \square \end{aligned}$$

4.3 Proofs

$\text{var}(\hat{\tau})$: We write $\hat{\tau}$ as

$$\hat{\tau} = \sum_{i=1}^n Z_i \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} \right\} - \frac{1}{n_0} \sum_{i=1}^n Y_i(0).$$

Then, the variance of $\hat{\tau}$ follows:

$$\begin{aligned} \text{var}(\hat{\tau}) &= \frac{n_1 n_0}{n(n-1)} \sum_{i=1}^n \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} - \frac{\bar{Y}(1)}{n_1} - \frac{\bar{Y}(0)}{n_0} \right\}^2 \\ &= \frac{n_1 n_0}{n(n-1)} \left[\frac{1}{n_1^2} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\}^2 + \frac{1}{n_0^2} \sum_{i=1}^n \{Y_i(0) - \bar{Y}(0)\}^2 \right. \\ &\quad \left. + \frac{2}{n_1 n_0} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\} \{Y_i(0) - \bar{Y}(0)\} \right] \\ &= \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1, 0). \end{aligned}$$

4.3 Proofs

Here, the first equality holds because of the following Lemma:

Lemma C.2

Under simple random sampling, the sample means are unbiased for the population means:

$$\mathbb{E} \left[\widehat{c} \right] = \bar{c}, \quad \mathbb{E} \left[\widehat{d} \right] = \bar{d}.$$

Their variances and covariance are

$$\text{var} \left(\widehat{c} \right) = \frac{n_0}{nn_1} S_c^2, \quad \text{var} \left(\widehat{d} \right) = \frac{n_0}{nn_1} S_d^2, \quad \text{cov} \left(\widehat{c}, \widehat{d} \right) = \frac{n_0}{nn_1} S_c d.$$

4.3 Proofs

Let $A_i = \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0}$, then $\text{var}(\hat{\tau}) = \text{var}(\sum_{i=1}^n Z_i A_i)$. In a CRE, we draw a simple random sample of size n_1 for a treatment group. Then, $\bar{A} = \frac{1}{n_1} \sum_{i=1}^n Z_i A_i$. Following Lemma C.2, $\text{var}(\bar{A}) = \frac{n_0}{nn_1} S_A^2$. Hence,

$$\text{var}(\hat{\tau}) = \text{var}\left(\sum_{i=1}^n Z_i A_i\right) = \text{var}(n_1 \bar{A}) = n_1^2 \text{var}(\bar{A}) = n_1^2 \frac{n_0}{nn_1} S_A^2 = \frac{n_0 n_1}{n} S_A^2.$$

Because $S_A^2 = \frac{1}{n-1} \sum_{i=1}^n \{A_i - \bar{A}\}^2$, we obtain

$$\text{var}(\hat{\tau}) = \frac{n_0 n_1}{n(n-1)} \sum_{i=1}^n \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} - \frac{\bar{Y}(1)}{n_1} - \frac{\bar{Y}(0)}{n_0} \right\}^2$$

4.3 Proofs

From Lemma 4.1, we also write the variance as

$$\text{var}(\hat{\tau}) = \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{1}{n} (S^2(1) + S^2(0) - S^2(\tau)) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}. \quad \square$$

4.4 Regression analysis of the CRE

- ▶ Practitioners often use regression-based inference for the average causal effect τ .
- ▶ A standard approach is OLS:

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{(a,b)} \sum_{i=1}^n (Y_i - a - bZ_i)^2.$$

- ▶ Here, $\hat{\tau} = \hat{\beta}$.
- ▶ However, the usual variance estimator from the OLS equals

$$\hat{V}_{\text{OLS}} = \frac{n(n_1 - 1)}{(n - 2)n_1 n_0} \hat{S}^2(1) + \frac{n(n_0 - 1)}{(n - 2)n_1 n_0} \hat{S}^2(0) \approx \frac{\hat{S}^2(1)}{n_0} + \frac{\hat{S}^2(0)}{n_1},$$

where the approximation holds with large n_1 and n_0 (it can differ).

- ▶ EHW robust variance estimator is closer to $\hat{V}$ and HC2 variant of the EHW robust variance estimator is identical to $\hat{V}$.

Neyman, J. (1923). On the application of probability theory to agricultural experiments. essay on principles. *Ann. Agricultural Sciences*, 1–51.